Matching the simulation parameters to the characterizing length scales

The various families of particles are characterized by different geometric parameters in the simulation programs. In this file, we introduce these parameters and describe how they map to the two scales σ and δ used in the corresponding manuscript.

Ellipses

The code for the ellipses uses the two semi-axes a and b as input. Here, a is the larger semi-axis and b is the smaller semi-axis. These two parameters specify $\sigma = 2b$ and $\delta = a/b - 1$.

Eggs

To determine the egg shape, the corresponding simulation code works with two parameters, a and b. In this case, a describes the larger semi-axis of the elliptical end of the particle and b corresponds to the radius of the semi-circular end of the particle as well as the shorter semi-axis of the elliptical end. We can calculate the parameters of the manuscript via $\sigma = 2b$ and $\delta = a/2b - 1/2$.

Lenses

For the lenses, the input parameters a and b describe the largest distance from the center to the surface and the shortest distance from the center to the surface, respectively. Therefore, they take the same role as the larger and smaller semi-axes of the ellipses, and we also find $\sigma = 2b$ and $\delta = a/b - 1$.

Stadia

The stadia simulation code takes the aspect ratio q as well as the diameter σ as input. This means that the length of the particle is given by $q\sigma$ and δ follows from $\delta = q - 1$.

Polygons

In the simulations, the polygon shape is determined by the number of vertices m, the length of the edges a, and a scaling parameter λ. Here, m and a characterize the regular representative, which is then stretched or compressed in one direction by the dimensionless factor λ. For particles with an odd number of vertices, the particles are stretched or compressed in a particle-fixed direction vector which points from the center to one of the vertices. For particles with an even number of vertices, the particles are stretched or compressed with respect to a direction that is perpendicular to one of the edges.

For $\lambda \geq 1$, the rescaling corresponds to stretching the particles in the horizontal direction with respect to the classification in the manuscript. Here, we have $\sigma = a/\sin(\pi/m)$ and $\delta = \lambda - 1$ (for the analyzed polygon families). The operation $\lambda < 1$ is equivalent to a stretching in the vertical direction when compared to the manuscript. Here, we have $\sigma = \lambda a/\sin(\pi/m)$ and $\delta = 1/\lambda - 1$ (for the analyzed polygon families).

Rhombi

The rhombi cannot be simulated with the standard polygon program. They have their separate code which takes the input parameters a and b. Note, however, that here a and b are the full long and full short diagonals, respectively. Consequently, $\sigma = b$ and $\delta = a/b - 1$.